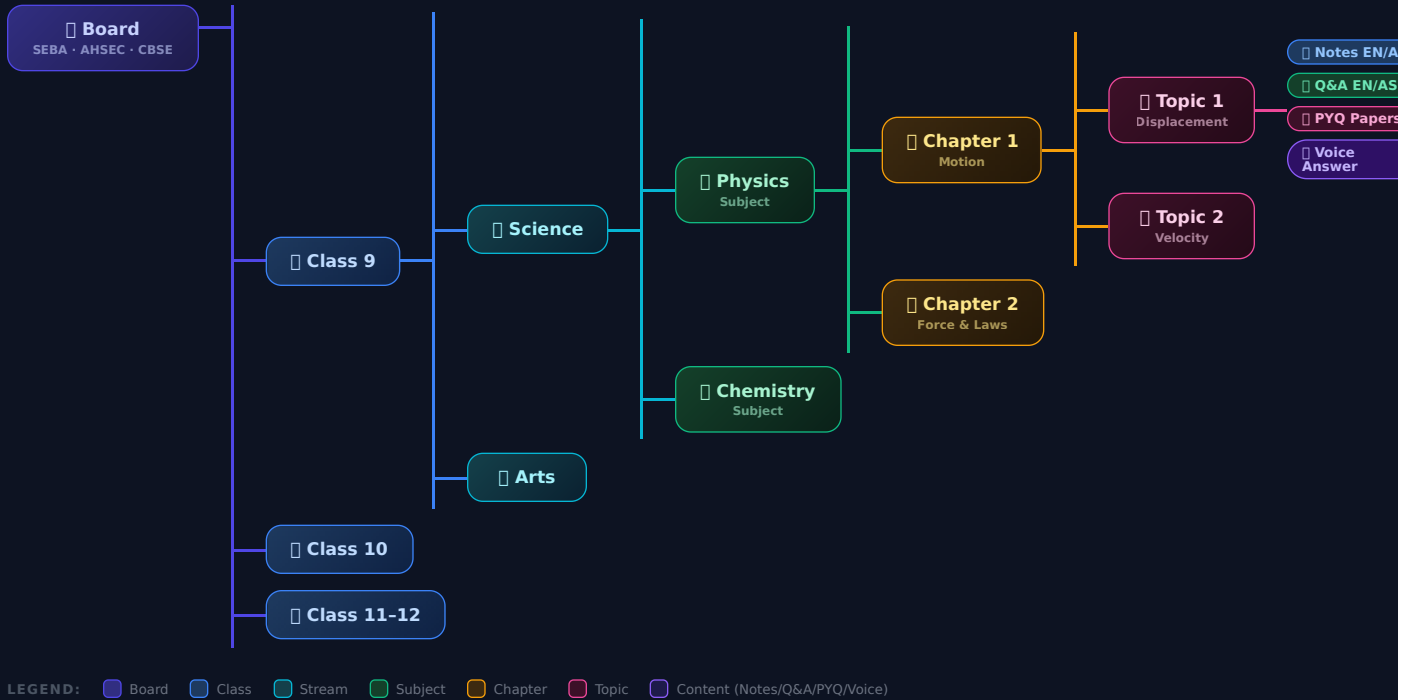


Syrabit.ai — Platform Architecture & What Makes It Novel

AI-Powered Bilingual Learning Assistant for SEBA & AHSEC Students · Northeast India's First Curriculum-Grounded EdTech AI

01 — SYLLABUS HIERARCHY IN LIBRARY PAGE



02 — CHAT PAGE · QUESTION → ANSWER PIPELINE (FULL MAPPING)



QUESTION → CURRICULUM MAPPING (RESOLVED AUTOMATICALLY)



Long-Term Memory Layer (unique to Syrabit)

Every conversation is stored per-student. Future queries are answered with awareness of what the student has already studied, their weak topics, and past doubts — creating a personalised tutor that improves over time.

03 — WHY SYRABIT VS. GENERIC AI (CHATGPT / GEMINI / BYJU'S)

FEATURE	SYRABIT.AI	CHATGPT / GENERIC AI
Assamese language support	Native Assamese text + voice via Sarvam AI — first in class	English-primary; Assamese responses are poor, unverified
Curriculum grounding (SEBA / AHSEC)	RAG over actual SEBA/AHSEC textbook content — zero hallucination on syllabus	General knowledge only; no awareness of Assam board curriculum
Syllabus-mapped answers	Every answer traceable to Board → Class → Subject → Chapter → Topic	No curriculum mapping; answers may cover wrong syllabus or board
Source citation per answer	Source card shows exact chapter & section used to generate answer	No citations; student cannot verify accuracy
Student memory & personalisation	Persistent per-student memory: weak topics, study history, past doubts	Stateless — forgets everything between sessions
Previous Year Questions (PYQ)	Structured PYQ database with solutions, searchable in chat	No exam-specific PYQ; training data cut-off, often outdated
Voice answer in Assamese	Sarvam TTS reads answers aloud in Assamese — accessibility for rural students	No Assamese TTS available
Cost for students	₹99-₹299/month (vs. ₹3,000+ coaching) — designed for affordability	ChatGPT Plus = \$20/month; no India pricing; coaching still needed
Designed for Assam / Northeast India	Built ground-up for SEBA/AHSEC; Assamese culture & language native	Built for global or Hindi-belt market; Northeast India is an afterthought
Staff / content moderation layer	Human-in-the-loop staff panel for curriculum quality control and content publishing	Fully automated; no subject-matter expert review of educational content

04 — FOUR INNOVATION PILLARS THAT DEFINE SYRABIT



VERNACULAR AI

First AI tutor to speak, read, and reason in Assamese. Sarvam AI multilingual models power both chat and voice — the 40+ lakh Assamese-medium students finally have an AI in their language.



CURRICULUM RAG

Retrieval-Augmented Generation over verified SEBA/AHSEC textbook content. Answers are pulled from the actual book, not hallucinated — every response is curriculum-exact and exam-safe.



STUDENT MEMORY

Long-term per-student memory tracks every studied topic, weak area, and past doubt. The tutor gets smarter about each student over time — no competitor offers this for Indian students.



FULL CONTENT STACK

Notes + Q&A + Previous Year Papers — one bilingual platform with human-level quality control. Eliminates the need for multiple apps, books, and coaching centres.

TECHNOLOGY:

Cloudflare Workers (Edge AI)

Cloudflare Vectorize (RAG)

Gemini 2.5 Flash (LLM)

Sarvam AI (Indic Language)

MongoDB Atlas (Database)

FastAPI + React (Fullstack)

BGE-M3 Embeddings (Multilingual)

Google Cloud Run (Deployment)